Jothi Baskar

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Education

Vellore Institute of Technology (VIT) — Vellore, India
B.Tech in Computer Science and Engineering (Specialisation in Internet of Things); CGPA: 8.43/10 — *2023 – 2027*

Narayana CO School — Bangalore, India
Higher Secondary School; Percentage: 89.20% — *2023*

Delhi Public School Bangalore East — Bangalore, India
Secondary School; Percentage: 93.17% — *2021*

Experience

Product / UX Intern — June 2026
axiTrust

- Analyzed axiTrust's B2B SaaS Insurer Portal, mapping how users navigate, filter, and manage applications.
- Designed Figma prototypes to simplify navigation, filtering, and application management for portal users.

Projects

AI-Powered Digital Twin for Smart Buildings | *Python, FastAPI, scikit-learn, SQLite, WebGL*

- Planned the project and designed its architecture around a simulated six-room, two-floor building, choosing the five sensor signals (temperature, humidity, occupancy, light, power) it monitors.
- Designed the interactive WebGL 3D twin embedded in the Streamlit dashboard, where rooms are coloured by status and can be inspected and controlled, so alerts and HVAC/lighting states are easy to locate.
- On simulated data, the system's rule engine (fed by Random Forest, Decision Tree, and Isolation Forest models) used about 5.5% less energy than a fixed HVAC schedule at similar comfort; all 45 automated tests passed.

IoT-Based Smart Parking System | *ESP8266, Embedded C/C++, MQTT, I2C* — GitHub

- In a three-member team, built a NodeMCU ESP8266 prototype that reads three IR sensors (two slots, one entry gate) and opens a servo gate only when a vehicle is waiting and a slot is free, so cars are not let into a full lot.
- Published slot occupancy, free-slot count, and gate state to MQTT over Wi-Fi via PubSubClient, so drivers can check availability remotely on an IoT MQTT Panel dashboard as well as on an on-site I2C LCD.

IoT-Based Smart Waste Segregation and Monitoring System | *Arduino, OpenCV, Flask, MQTT* — GitHub

- Built the hardware for a three-member team's cardboard sorter (servo flaps, metal detector, IR sensor) and wrote Arduino firmware that drives each flap from a host OpenCV/scikit-learn classifier (paper, metal, organic).
- Extended it into a separate Flask/SQLite/MQTT software version with rule-based wet/dry/metal sorting and fill-level alerts that flag bins nearing capacity for collection.

Smart LPG Leak Detection & Automatic Shut-Off System | *Tinkercad, Python, FastAPI, MQTT* — GitHub

- Prototyped the circuit in Tinkercad, then built a FastAPI simulation of MQ-5/MQ-6 data across seven leak and fault scenarios, using EMA filtering and a rate-of-rise check to flag fast leaks before fixed thresholds are reached.
- Implemented a four-level risk engine with hysteresis and sensor-fault diagnostics that latches a simulated servo valve shut until an operator resets it, with SQLite event logging and MQTT publishing.

Fog-Assisted Real-Time Shuttle Tracking System | *Python, Flask, REST APIs, JavaScript* — GitHub

- Designed a three-tier Edge–Fog–Cloud architecture with Flask REST services, computing route-aware ETAs on a local fog node from Haversine distances along campus route legs for per-stop arrival estimates.
- Built an SVG digital-twin dashboard that snaps shuttles to route segments, giving a live view of shuttle positions.

Technical Skills

Languages: C, Embedded C, C++, Python, Java, JavaScript, HTML/CSS
CS Fundamentals: DSA, OOP, OS, Computer Networks, DBMS, Computer Architecture, AI
Embedded Systems: Arduino, ESP8266, Microcontrollers, Digital Systems Design, Sensors and Actuators, UART, I2C
IoT & Networking: IoT Architectures and Protocols, MQTT, REST APIs, Edge and Fog Computing

Certifications

NVIDIA – Fundamentals of Deep Learning — *September 2026*